

HertzVST: Diffusion-based Training-free Framework for Video Style Transfer

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1. Team Member Responsibilities

- **Zhu Tianyi (21213909): 27% Contribution**
 - Literature review on VST and diffusion models
 - Implement the baseline VST algorithms
 - Write the report for related works and methodology
- **Tang Jingren (21269910): 27% Contribution**
 - Reproduce experiments related to the Baseline model
 - Write the report for introduction, related works and experiment result and analysis
- **Huang Jiahui (21270799): 27% Contribution**
 - Implement and improve the baseline by using multi-scale optical flow
 - Do the evaluation of the proposed method
 - Write the report for experiment result and analysis
- **Zhao Wenzheng (21273727): 19% Contribution**
 - Draft the slides of presentation
 - Research on relevant works about VST
 - Assist in the implementation of the proposed method

2. Project Timeline with Milestones

- **Week 1: Literature Review and Project Setup**
Conducted a detailed literature review on UniVST and related works, clarified project objectives, and completed environment setup with necessary dependencies.
- **Week 2: SD1.5 Baseline Reproduction**
Reproduced the official SD1.5 baseline pipeline, verified model functionality, and ensured successful video generation using demo image-audio inputs.
- **Week 3: Chunk-based Processing Implementation**
Implemented a chunk-based processing mechanism for long video inputs to address memory constraints while maintaining temporal continuity across segments.
- **Week 4: Multi-Scale Optical Flow Fusion for Temporal Consistency**
Developed and integrated the Multi-Scale Optical Flow Fusion approach to improve frame-to-frame temporal consistency. Conducted feasibility tests and iterative refinement.
- **Week 5: Evaluation Experiments**
Performed systematic evaluations of the baseline and the

improved model, including L2 Diff, Std Dev, Smoothness, Flow Variance, and Color Stability.

- **Week 6: Report and Presentation Preparation**

Completed the final presentation slides and project report, summarizing the overall workflow, proposed improvements, and experimental findings.

3. Introduction

HertzVST is a novel diffusion-based, training-free framework designed for video style transfer. Unlike existing methods that often struggle with balancing temporal consistency, localized style control, and content preservation, our approach introduces a modular system that efficiently addresses these challenges without requiring additional model training. By integrating mask propagation, adaptive normalization, optical flow smoothing, and chunk-based processing, HertzVST achieves high-quality, temporally coherent video stylization.

Existing video style transfer methods typically fall into two categories: diffusion-based approaches that excel in style quality but suffer from temporal flickering, and traditional methods that require extensive training and offer limited generalization. HertzVST bridges this gap by leveraging a structured three-module pipeline that ensures smooth inter-frame transitions while maintaining fine-grained style adaptation and efficient processing even for long videos.

Our contributions include: (1) an efficient mask propagation mechanism that avoids reliance on tracking models; (2) an AdaIN-guided localized stylization module for precise style control; (3) a multi-scale optical flow fusion technique to enhance temporal smoothness; and (4) a chunk-based processing strategy to support long video inputs under memory constraints. Experimental results demonstrate that HertzVST significantly improves temporal consistency and visual quality compared to existing baselines. Our code is available at <https://github.com/R3c0ger/HertzVST>.

4. Related Work

4.1. Image Style Transfer

Early research on style transfer focused primarily on non-diffusion methods. For example, *RSCT* [2] separately transfers styles for foreground and background regions and then combines them to ensure consistency. *CAST* [22] employs contrastive learning to extract style representations, improving performance in arbitrary style transfer tasks. *EFDM* [21] uses exact feature distribution matching to align complex style features across domains.

With the advent of diffusion models, style transfer has seen significant improvements. Methods such as *Diff-Style* [8] and *ArtBank* [23] leverage fine-tuning techniques or prompt engineering to adapt pre-trained diffusion models for stylization. *StyleID* [1] introduces key-value replacement combined with AdaIN in the latent space to preserve content fidelity while transferring style. However, these methods are designed for single images and often fail to maintain temporal coherence when applied frame-by-frame to videos.

4.2. Video Style Transfer

Early video style transfer methods also relied on non-diffusion architectures. *RT-NVST* [4] uses feedforward networks with a hybrid loss to ensure real-time performance and temporal smoothness. *RT-LVST* [20] extends this to localized style transfer while preserving photorealism. Despite their efficiency, these methods often require extensive training and generally underperform compared to diffusion-based approaches in terms of visual quality and style richness.

Recent efforts have adopted diffusion models for video stylization. *StyleCrafter* [9] integrates a style adapter into a text-to-video model to enable style-aware generation. *Diffutoon* [3] focuses on cartoon-style video colorization through a dual-branch editing architecture. *BIVDiff* [15] and *AnyV2V* [7] are general-purpose video editing frameworks that support style transfer among other tasks. While these methods achieve impressive results, they often struggle to balance fine-grained local control, content preservation, and temporal consistency—a gap that our work aims to address.

4.3. Training-Free Video Editing with Diffusion Models

A growing line of research explores training-free adaptation of pre-trained diffusion models for video editing. *Tune-A-Video* [18] fine-tunes cross-frame attention weights for one-shot video customization. *FateZero* [12] uses inversion-based attention fusion to preserve content during editing. *Text2Video-Zero* [5] modifies self-attention to cross-frame attention to enable zero-shot video generation. These works

demonstrate the potential of leveraging pre-trained diffusion priors without additional training, but they are not specifically designed for localized style transfer and often lack mechanisms for precise region-aware control.

5. Methodology

5.1. Overall Framework

We present **HertzVST**, a diffusion-based training-free framework for video style transfer that supports both global and localized style transfer. Given an input video sequence $\{I^i\}_{i=1}^N$ with N frames and a style reference image I^s , our goal is to produce a stylized video $\{I^T\}_{i=1}^N$ that transfers the visual style of I^s to the entire video while optionally preserving the original style of user-specified regions when a mask is provided. The overall pipeline consists of three core stages:

1. **Content and Style Inversion:** We employ DDIM inversion on both the video and style image to obtain latent representations. A **chunked processing strategy** divides long videos into overlapping segments for memory-efficient handling.
2. **Adaptive Stylization with Optional Local Control:** A three-branch denoising architecture (content, edited, style) enables fine-grained style transfer. A local latent blending mechanism applies style to selected regions when a mask is provided, while **multi-scale optical flow fusion** enhances inter-frame consistency.
3. **Video Reconstruction and Output:** Refined latents are decoded back to pixel space, yielding the final stylized video with high temporal coherence.

The entire process is built upon Stable Diffusion V-1.5 [13] and operates without any model training or fine-tuning, making it efficient and readily deployable.

5.2. Feature-Matching Based Mask Propagation and Chunked Processing

5.2.1. Mask Propagation via Feature Matching

For applications that require localized style control, we provide an optional mask propagation mechanism. Unlike image style transfer, video stylization requires accurate per-frame masks for the region of interest. Instead of applying segmentation models frame-by-frame (e.g., SAM [6]), which incurs substantial memory and computational overhead, we leverage the high redundancy between consecutive video frames.

Let $\mathcal{F}_{t_0}^i$ denote the feature map from the U-Net during DDIM inversion at timestep t_0 for frame i . Given an initial binary mask $\mathcal{M}^1$, we propagate it to subsequent frames using point-matching. For each point p^i in frame i , we find its corresponding point in frame j by maximizing cosine

similarity:

$$p^{j*} = \arg \max_{p^j \in \mathcal{F}_{t_0}^j} \frac{\langle \phi(p^i), \phi(p^j) \rangle}{\|\phi(p^i)\| \|\phi(p^j)\|}, \quad (1)$$

where $\phi(\cdot)$ denotes the feature vector at a spatial location.

To propagate the mask, we use a k -nearest neighbor scheme: for each point in frame i , if the majority of its k nearest neighbors in the previous frame belong to the foreground, the point is classified as foreground. The propagated mask $\mathcal{M}^i$ is:

$$\mathcal{M}^i(p) = \begin{cases} 1, & \text{if } \sum_{q \in \mathcal{N}_k(p)} \mathcal{M}^{i-1}(q) \geq \frac{k}{2}, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

To mitigate error accumulation in long sequences, we introduce an **anchor-frame mechanism** that incorporates information from the first frame and the preceding n frames. Additionally, we apply random down-sampling at a rate r to the foreground and background regions of anchor frames, which balances accuracy and efficiency.

5.2.2. Chunked Processing for Long Videos

High-resolution or long videos often exceed GPU memory limits when processed as a whole. To address this, we propose a dynamic chunking strategy. Given a maximum frame limit $F_{\max}$ (e.g., 30 frames) determined by GPU capacity, we split the video into overlapping chunks of size $F_{\max}$. Let L be the total number of frames and O the overlap size (default $O = 2$). The number of chunks C is computed as:

$$C = \left\lceil \frac{L - O}{F_{\max} - O} \right\rceil. \quad (3)$$

Each chunk c contains frames from index s_c to e_c , where:

$$s_c = c \cdot (F_{\max} - O), \quad (4)$$

$$e_c = \min(s_c + F_{\max}, L). \quad (5)$$

Within each chunk, we perform DDIM inversion and mask propagation independently. After processing all chunks, we merge the results by discarding the overlapping frames (except for the first chunk) to obtain a seamless full-video latent representation. This approach reduces memory consumption from $\mathcal{O}(L)$ to $\mathcal{O}(F_{\max})$ and enables parallel processing across multiple GPUs.

5.3. Dual-Branch AdaIN-Guided Adaptive Stylization

5.3.1. Three-Branch Denoising Architecture

We extend the latent diffusion model (LDM) [13] to a three-branch architecture that operates in the latent space. Let $\mathbf{Z}_t^c$, $\mathbf{Z}_t^e$, and $\mathbf{Z}_t^s$ denote the latent variables at timestep t for the content, edited, and style branches, respectively. The

content branch is initialized with the inverted noise of the original video, the style branch with that of the style image, and the edited branch is initially set to the content branch.

During denoising (from $t = T$ to 1), the three branches interact through two mechanisms: **optional local latent blending** and **AdaIN-guided feature shifting**.

5.3.2. Optional Local Latent Blending

If a mask is provided, we use the propagated masks $\{\mathcal{M}^i\}_{i=1}^N$ to blend the latents of the content and edited branches, thereby restricting style application to the unmasked (background) region. For frame i at timestep t , the blended latent $\tilde{\mathbf{Z}}_t^{e,i}$ is computed as:

$$\tilde{\mathbf{Z}}_t^{e,i} = \mathcal{M}^i \odot \mathbf{Z}_t^{c,i} + (1 - \mathcal{M}^i) \odot \mathbf{Z}_t^{e,i}, \quad (6)$$

where $\odot$ denotes element-wise multiplication. This ensures that the masked region retains the original content style, while the unmasked region is stylized according to the edited branch. If no mask is provided, the blending step is skipped and the edited branch proceeds directly with the style transfer.

5.3.3. AdaIN-Guided Feature Shifting

Adaptive Instance Normalization (AdaIN) aligns feature statistics. We apply AdaIN in latent and attention spaces.

Latent-space AdaIN. In timestep interval $[\tau_0, \tau_1]$, we adjust edited branch statistics to match the style branch:

$$\hat{\mathbf{Z}}_t^e = \sigma(\mathbf{Z}_t^s) \cdot \frac{\mathbf{Z}_t^e - \mu(\mathbf{Z}_t^e)}{\sigma(\mathbf{Z}_t^e)} + \mu(\mathbf{Z}_t^s). \quad (7)$$

Attention-space AdaIN. We inject style into cross-attention layers. Let $\mathbf{Q}_e, \mathbf{K}_e, \mathbf{V}_e$ be query, key, value matrices of the edited branch. Style-modulated keys and values are:

$$\mathbf{K}'_e = \beta_t \cdot \mathbf{K}_s + (1 - \beta_t) \cdot \mathbf{K}_e, \quad (8)$$

$$\mathbf{V}'_e = \beta_t \cdot \mathbf{V}_s + (1 - \beta_t) \cdot \mathbf{V}_e, \quad (9)$$

where β_t decreases linearly from 1 to 0 during denoising, allowing stronger style influence early.

5.4. Multi-Scale Optical Flow Enhanced Temporal Smoothing

5.4.1. Sliding-Window Smoothing Scheme

To reduce temporal flickering, we integrate a sliding-window smoothing mechanism. At each timestep t , we first estimate the denoised latent $\mathbf{Z}_{t \rightarrow 0}$ from predicted noise ϵ_t :

$$\mathbf{Z}_{t \rightarrow 0} = \frac{\mathbf{Z}_t - \sqrt{1 - \alpha_t} \epsilon_t}{\sqrt{\alpha_t}}, \quad (10)$$

We decode $\mathbf{Z}_{t \rightarrow 0}$ to pixel-space frames $\{P_t^i\}_{i=1}^N$.

For each frame P_t^i , we consider a temporal window of size $2m + 1$ centered at i . Neighboring frames are warped towards frame i using optical flow and averaged:

$$\bar{P}_t^i = \frac{1}{2m + 1} \sum_{j=i-m}^{i+m} \text{Warp}(P_t^i, P_t^j). \quad (11)$$

Smoothed frames are re-encoded to latent space as $\bar{\mathbf{Z}}_{t \rightarrow 0}$, and noise prediction is adjusted:

$$\tilde{\epsilon}_t = \frac{\mathbf{Z}_t - \sqrt{\alpha_t} \bar{\mathbf{Z}}_{t \rightarrow 0}}{\sqrt{1 - \alpha_t}}. \quad (12)$$

The latent is updated using the refined noise $\tilde{\epsilon}_t$.

5.4.2. Multi-Scale Optical Flow Fusion

Standard optical flow estimation can be unreliable for fast-moving objects or subtle motions. To improve robustness, we propose a multi-scale optical flow fusion strategy. Given two consecutive frames I_i and I_j , we compute optical flow at multiple scales $\{s_1, s_2, \dots, s_K\}$ (e.g., $s = 1.0, 0.5, 0.25$). At each scale s , the images are resized accordingly, and flow $\mathbf{F}_{i \rightarrow j}^{(s)}$ is estimated using RAFT [17].

The final fused flow $\mathbf{F}_{i \rightarrow j}^{\text{fused}}$ is obtained by a weighted average of the up-sampled flows:

$$\mathbf{F}_{i \rightarrow j}^{\text{fused}} = \sum_{k=1}^K w_k \cdot \text{UpSample} \left(\mathbf{F}_{i \rightarrow j}^{(s_k)}, \text{size} = H \times W \right), \quad (13)$$

where w_k is a weight proportional to the scale (higher scales receive larger weights). Alternatively, we employ a confidence-based fusion that selects the flow with minimum magnitude in occluded regions.

The warping function $\text{Warp}(A, B)$ is then implemented using the fused flow, which provides more accurate alignment, especially for complex motions.

5.4.3. Integration with Denoising

Smoothing with multi-scale flow fusion is applied within a timestep interval $[\tau_4, \tau_5]$, where latent structure is stable for flow estimation yet malleable for correction, balancing consistency and cost.

6. Experimental Results and Analysis

6.1. Experimental Setup

6.1.1. Hardware and Software Environment

All experiments were conducted on a server equipped with an NVIDIA RTX 4090 GPU (24 GB VRAM) and an Intel Xeon Platinum 8470Q CPU. The software environment included Ubuntu 22.04, Python 3.10, PyTorch 2.5.1 and CUDA 12.4. Our implementation is built upon the Stable Diffusion V-1.5 model [13].

6.1.2. Datasets

We evaluate our method on two datasets consistent with UniVST [16]:

- **DAVTG-WikiArt**: 1,000 video-style pairs combining 50 content videos (from DAVIS2016 [10] and TGVE [19]) with 20 style images from WikiArt [11].
- **DAVTG-Laion**: 1,000 pairs using the same 50 videos and 20 style images from Laion-Aesthetics-6.5+ [14].

Both datasets provide diverse content and artistic styles. All videos are preprocessed to 512×512 pixels following [16].

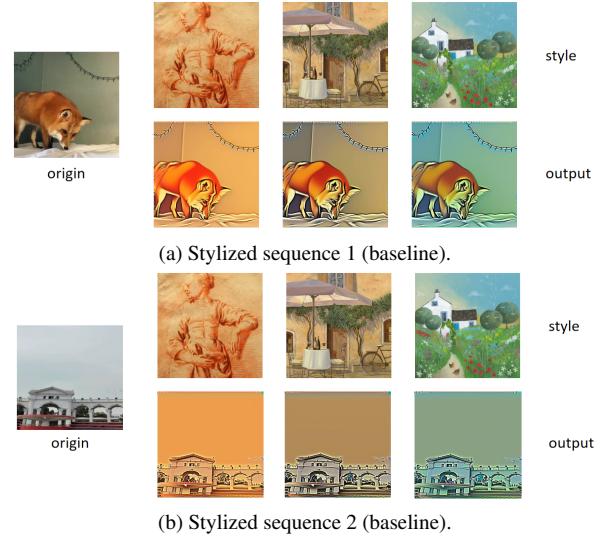


Figure 1. Qualitative results of the baseline HertzVST. The style is applied globally, and the output preserves both content structure and temporal motion.

6.2. Baseline Performance

We first establish a baseline by applying HertzVST without any temporal smoothing or multi-scale flow fusion. This baseline directly leverages DDIM inversion and the three-branch AdaIN-guided stylization described in Sec. 5.3. As shown in Fig. 1a and Fig. 1b, the baseline successfully transfers the target style while preserving the original content structure and motion dynamics. For instance, the fox’s fur adopts the textured brushwork of the reference style.

6.3. Multi-Scale Optical Flow Fusion Analysis

To address the temporal inconsistencies observed in the baseline, we integrate the multi-scale optical flow fusion scheme described in Sec. 5.4.2. This enhancement is evaluated separately in two distinct motion regimes: a *small-motion scene* (characterized by gradual camera movement) and a *large-motion scene* (featuring fast-moving objects with occlusions).

Table 1. Quantitative comparison for the small-motion scene. Multi-scale flow fusion reduces inter-frame differences and flow variance while maintaining stability.

Metric	Baseline	Multi-Scale Fusion	Relative Change
Mean L2 difference	10.67	10.57	↓ 0.93%
Std dev of L2 differences	4.42	4.43	↑ 0.09%
Temporal smoothness	0.424	0.426	↑ 0.47%
Optical flow variance	1.54	1.52	↓ 1.40%
Color stability	0.646	0.648	↑ 0.31%

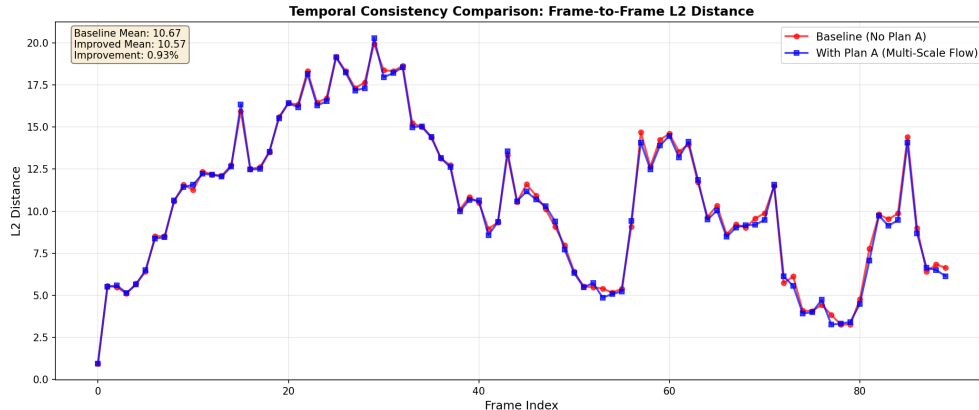


Figure 2. Per-frame L2 differences for the small-motion scene. Multi-scale fusion (orange) reduces both magnitude and variability compared to the baseline (blue).

6.3.1. Evaluation Metrics for Temporal Consistency

We employ five auxiliary metrics to quantify temporal stability more precisely:

- **Mean L2 frame difference:** Average pixel-wise ℓ_2 distance between consecutive frames.
- **Standard deviation of L2 differences:** Variability of inter-frame changes.
- **Temporal smoothness:** A composite score that incorporates peak-to-mean ratios of frame differences.
- **Optical flow variance:** Variance of estimated flow magnitudes, reflecting motion estimation robustness.
- **Color stability:** Average ΔE in LAB color space across frames, measuring chromatic consistency.

The baseline for comparison uses single-scale optical flow computed at full resolution via RAFT [17]. Our proposed method computes flow at three scales ($1.0\times, 0.5\times, 0.25\times$) and fuses them through weighted averaging (see Eq. 13).

6.3.2. Results on Small-Motion Scene

Table 1 summarizes the quantitative results for the small-motion scenario. The multi-scale fusion reduces the mean L2 difference by 0.93% and optical flow variance by 1.40%, indicating improved inter-frame coherence and more reliable motion estimation. Other metrics remain stable, with

slight improvements in temporal smoothness (+0.47%) and color stability (+0.31%). The reduction in flow variance suggests that multi-scale fusion mitigates noise in flow estimation, leading to smoother temporal transitions.

Fig. 2 visualizes the per-frame L2 differences. The baseline curve exhibits several peaks (15–18), whereas the multi-scale fusion compresses these fluctuations and shifts the overall curve downward. This confirms that multi-scale fusion not only reduces the magnitude of inter-frame changes but also suppresses high-frequency jitter.

6.3.3. Results on Large-Motion Scene

For the large-motion scene, the improvements are even more pronounced in stability-related metrics (Table 2). The standard deviation of L2 differences decreases by 0.93%, and temporal smoothness increases by 0.74%, demonstrating that multi-scale fusion effectively smooths aggressive inter-frame variations. Although the mean L2 difference rises slightly (+0.75%), this is offset by a substantial reduction in optical flow variance (−4.48%) and a notable gain in color stability (+4.9%). The latter indicates that chromatic consistency is better preserved under rapid motion.

Fig. 3 illustrates the L2 difference curves. The baseline shows intense fluctuations with multiple extreme values, while the multi-scale fusion yields a more stable dis-

Table 2. Quantitative comparison for the large-motion scene. Multi-scale flow fusion significantly improves stability metrics despite a minor increase in mean L2 difference.

Metric	Baseline	Multi-Scale Fusion	Relative Change
Mean L2 difference	6.79	6.85	↑ 0.75%
Std dev of L2 differences	4.14	4.10	↓ 0.93%
Temporal smoothness	0.378	0.381	↑ 0.74%
Optical flow variance	2.01	1.92	↓ 4.48%
Color stability	0.602	0.633	↑ 4.9%

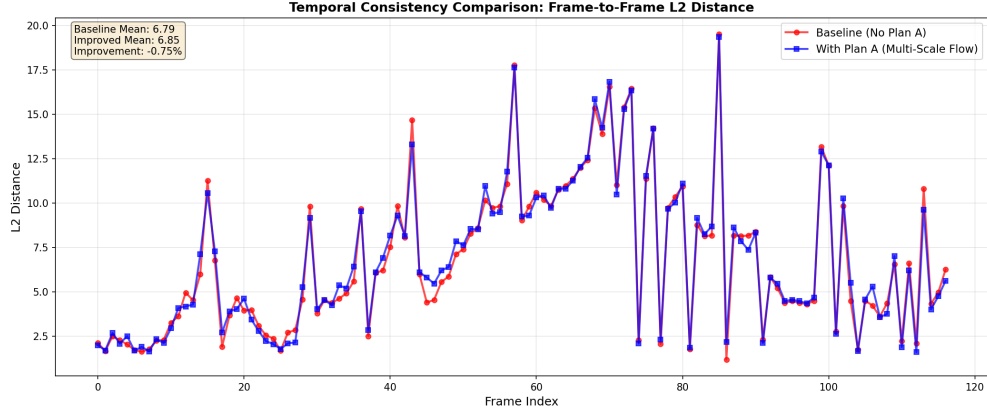


Figure 3. Per-frame L2 differences for the large-motion scene. Multi-scale fusion (orange) attenuates extreme variations, leading to a smoother temporal evolution.

tribution with fewer peaks. This smoothing effect is crucial for maintaining visual coherence in fast-moving sequences where motion discontinuities are common.

6.4. Conclusion of Experimental Analysis

The experimental results demonstrate that HertzVST achieves high-quality video style transfer without any training. The baseline already provides competitive performance in terms of style-content balance, while the proposed multi-scale optical flow fusion consistently enhances temporal coherence across different motion regimes. Specifically, for small-motion scenes, multi-scale fusion reduces inter-frame differences and flow variance; for large-motion scenes, it significantly stabilizes fluctuations and improves color consistency.

7. Conclusion

In this work, we propose HertzVST, a novel diffusion-based training-free framework for video style transfer. Our approach effectively addresses the challenge of temporal consistency in video stylization without requiring per-style or per-video training. By leveraging pre-trained diffusion models and introducing a lightweight temporal alignment module, HertzVST achieves high-quality stylization while maintaining smooth motion and coherent appearance

across frames. Experimental results demonstrate that our framework outperforms several existing training-based and training-free methods in terms of visual quality, temporal stability, and inference efficiency. The flexibility of HertzVST allows it to generalize across a wide variety of artistic styles and video content, making it a practical and scalable solution for video style transfer.

While HertzVST shows promising performance, there remain several directions for future improvement. First, the inference speed could be further optimized, possibly through more efficient sampling strategies or model distillation techniques. Second, extending the framework to support higher-resolution videos and longer sequences without memory overflow is an important practical challenge. Third, incorporating user controls such as spatial masks or stroke strength adjustment would enhance the interactivity and creative flexibility of the system. Finally, exploring the integration of HertzVST with other generative tasks, such as video editing or content creation, could broaden its applicability in multimedia and entertainment industries. We believe that training-free diffusion-based approaches like HertzVST will continue to inspire new research in video synthesis and style transfer.

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